Report to document the development of the Intelligent system using both bottom-up and top-down approaches should include the following:

1.Compare the advantages and challenges of several tools and techniques for the development of Intelligent Systems. (JOHNSTON, 2024) (Unveiling the Tools of AI: Techniques for Intelligent Systems, 2024)

Tools	***Advantages***	***Challenges***
TensorFlow	Highly flexible, Open-source platform anyone can use it, scalability not limited to one specific device, compatible to many programming languages allowing a large scale of user to use it make it flexible to use. Very efficient in Deep learning unlike scikit- Learn.	Frequents updates requires update from the user from time to time. No GPU support for Nvidia and only language support It requires fundamental knowledge of advanced calculus and linear algebra along with a good understanding of machine learning can be complex for beginners.
Microsoft Azure ML	Cloud-based storage this help the user to not to buy a high computational power and large amount of memory computer to work on the dataset , Global Reach and Reliability gives user a seamless experience regardless on their location, Robust Security and Compliance Azure known for its advance security and certificates in compliance	Requires subscription which prevent user to use, dependent on internet, limited storage on the free trail, a vast of feature and options can be overwhelming for new users on how to use them(steep learning curve)
Scikit-Learn	Open source and commercially used, easy to use very user friendly since its easy to import using python platform, in addition its contains supports a wide range of ML algorithm and processing tools also integrates well with other	Limited deep learning capabilities and run slows on large dataset also cannot handle high dimensionality data

	libraries like Pandas and NumPy.	

Techniques	***Advantages***	***Challenges***
Supervised Learning	High accuracy, well-defined output	Requires labeled data, risk of overfitting or underfitting
Unsupervised Learning	Identifies hidden patterns, no need for labeled data	Less interpretable, results can be unpredictable
Reinforcement Learning	Adapts to dynamic environments, learns from experience	Requires extensive training, computationally expensive

2. investigate options around the approaches, techniques and tools for the deployment of modern Intelligent Systems.

Investigating the options to deploy a modern intelligent system effectively, can be done by exploring a range of tools, techniques, and deployment options, starting with **Tools** such as TensorFlow, Scikit-learn, etc. Let us start with TensorFlow, TensorFlow offers amazing libraries and frameworks to use for developing machine learning models efficiently also very efficient, especially for large-scale neural network data helping us in training and deploying them. In addition, Scikit-learn is very suitable for machine learning algorithms since it helps the user work well with simple algorithm such as pandas, NumPy, etc. For deployment, we can use platforms like **Microsoft Azure AI** provide end-to-end solutions providing the user powerful tools that can help them create innovative solutions using machine learning, natural language processing (NLP), and computer vision and all of these can be done by its services like Azure Machine Learning service.

Now let's dig into the techniques for the deployment of modern Intelligent Systems, intelligent systems can be built using supervised learning, unsupervised learning, or reinforcement learning we choose what kind of technique to use depending on the problem that we have and the dataset of that problem. Supervised learning is very effective for tasks that contain labelled data such as sentiment analysis, image classification, etc. However, unsupervised learning works by identifying hidden patterns or groupings in data because the data that we have isn’t labelled data such as clustering techniques. Reinforcement learning is unlike the others it involves learning without relying on a dataset it involves learning through experience. In Reinforcement learning agents learn optimal actions by interacting with their environment is done by performing actions and receiving feedback, like robotics or game AI.

Last but not least, for deployment we can use various cloud-based solutions for example Azure, AWS, or Google Cloud. cloud platforms can host intelligent systems by using its platform which is Azure Machine Learning which holds various features and functionalities essential for data scientists or even a program that facilitates several tasks such as data preparation, model training, predictive service deployment, and usage monitoring in the top of all of that it allows us to deploy machine learning models without the need for coding. These services ensure that models are available by application programming interfaces that are known for APIs example of the interface is the interface between web and mobile applications that allow us to send user data such as age, height, weight and more. this helps us to collect data for the machine learning

model hosted on the cloud. Additionally, edge computing allows the models to run directly on devices like smartphones, IoT sensors, or smart appliances, reducing latency and ensuring data privacy. This hybrid approach of cloud and edge deployment boosts performance and scalability and enables intelligent systems to cater to diverse environments and user requirements.

The **top-down** and **bottom-up** approaches are strategies used in problem-solving, software development, and system design

Let's define them:

the top-down approach goes from the general to the specific which means it start by high level over view to a low level breaking it into smaller sub problems (e.g., classify emails or predict prices), and the bottom-up approach begins at the specific and moves to the general it start smaller and build upwards to creates a full system (e.g., data cleaning, basic algorithms)

Bottom-Up: specific tools, techniques, and local deployment methods, leading to their integration into larger systems.

Top-Down: cloud-based solutions, learning methods, and hybrid deployment strategies, then moved into specific tools and applications that exemplify these ideas.

The top down and bottom-up approaches that we mentioned:

Tools for Deployment:

- **Starting with TensorFlow**
- **Scikit-learn**: Pandas and NumPy.

Techniques for Intelligent Systems:

- individual techniques—**supervised learning**, **unsupervised learning**, and **reinforcement learning**—e.g., supervised learning for labeled data, reinforcement learning for environment-based optimization)

Edge Computing:

- TensorFlow Lite on IoT devices and smart appliances, to showcase how models can perform localized tasks effectively.

Top-Down Approach (Starting with the Big Picture to Details):

Tools for Deployment:

- **cloud platforms** like Microsoft Azure AI.
- cloud services such as APIs applications like web and mobile systems.

Techniques for Intelligent Systems:

- unsupervised, and reinforcement learning high-level applications like sentiment analysis, clustering.

Hybrid Deployment Strategies:

- **cloud and edge computing**,

3. Demonstrate how different approaches and tools work together for the deployment of Intelligent System.

We started with using a machine learning algorithm, Logistic regression was chosen as the machine learning algorithm because it's used for classification data where the goal is to predict a value based on the features that we have in the scenario that we had we needed to predict how much is the risk for the person to have diabetes based on input features (e.g., BMI, HbA1c) predicting output (diabetes risk) this will provide insights into the relationship features and the target feature making it suitable for diabetes medical application where explainability is critical. Additionally, logistic regression can handle a large scale of data making it very efficient to use since it provides a probabilistic output.

Fuzzy logic was incorporated to handle uncertainty and imprecision in medical data since the medical data might contain noise or be incomplete one fuzzy logic helps us to solve this issue by putting it in partial truths (e.g., blood glucose is "slightly high" or "moderately) and the advantage of this truth we can differentiate these symptoms level or diabetes level using degree ('High', 'Very High', 'Low') after we bring input features and use rules on them such as IF blood glucose is high AND BMI is obese, THEN risk of diabetes is high, which is near to human-like reasoning that combines multiple inputs into one output. fuzzy logic uses rules for example "if BMI is high and HbA1c is high, then diabetes risk is high" In this way using these rules is called human-like reasoning that is easy to understand and aligns with how medical professionals would think. Combining logistic regression and fuzzy logic is a hybrid approach that will improve model generalization by combining human-like reasoning (fuzzy logic) with statistical output predictions (from logistic regression) also handles uncertainty and improves interpretability.

As for deploying diabetes intelligent system we used azure since it excels comparing to the others by providing a set of tools service for data management, application hosting and even model training which very useful for the data scientist and programmer. We used Azure machine learning service that Azure have to train and optimize the logistic model and the fuzzy logic they are implemented on the system as serverless compute service providing lake of storage that highly secured by Microsoft company using strong encryptions it also tracks the system performance with providing error if any issue occurred. In addition, azure have an app service makes it accessible to end-users enabling doctors and patients to have interface connecting them with each other collecting the data from the patient and displaying the result of the diabetes risk for this patient.

To conclude the combination of logistic regression and fuzzy logic, supported by Azure's scalable and secure infrastructure that is protected by Microsoft company creates a diabetes prediction system that is both accurate and interpretable and used to help the user to know how much the risk of him getting diabetes. This approach is particularly suited for healthcare applications since we have provided and achieved transparency, scalability, and robustness these are very crucial to have on any intelligent system. Moreover, Azure has helped the system to handle growing data and user demands while maintaining compliance with healthcare

regulations. Overall, all of these integrations we mentioned demonstrate how different approaches and tools can work together to deploy intelligent systems effectively.

Summary of the steps we talked about

Step 1: Data Collection

- Patient data (e.g., BMI, HbA1c, age, blood glucose) is collected and stored in **Azure Data Lake Storage**.

Step 2: Model Training

- The logistic regression model is trained using **Azure Machine Learning** on historical patient data.

Step 3: Prediction

- For a new patient, the logistic regression model predicts the probability of diabetes (e.g., 0.6 for high risk).

Step 4: Fuzzy Logic

- The probability and other features (e.g., age, blood glucose) are passed to the **Azure *Function*** *implementing fuzzy logic.*
- *Fuzzy rules are applied to refine the risk level (e.g., "high", "medium", "low").*

Step 5: Deployment

- *The final risk level is displayed to the user via a web application hosted on **Azure App Service**.*

Step 6: Monitoring

- ***Azure Monitor** tracks the system's performance and alerts the team in case of issues.*

.Example Use Case

Input:

- *BMI = 28.5, HbA1c = 6.7, Age = 45, Blood Glucose = 120*

Process:

1. *Logistic regression predicts a probability of 0.6 (high risk).*
2. *Fuzzy logic refines the risk level to "high" based on rules.*

Output:

- *Diabetes Risk: High*

4.Evaluate the potential impact of deploying several types, approaches and tools of AI and Intelligent Systems on both users and organizations. (Vohra, 2024) (Dippnall, n.d.) (Challenges with developing and deploying AI models and applications in industrial systems, 2024) (Artificial intelligence, machine learning and deep learning in advanced robotics, a review, n.d.) (Maliugina, 2024) (Levy, 2024)

Potential Impact of Deploying AI and Intelligent Systems

1. ***Impact on Users:***

- *Enhanced User Experience:*

  Analyzing the user behavior and preferences by providing him personalized recommendations and service to what he likes this will increase customer satisfaction such as automation of routine task like chatbots, virtual assistant that will increase the user expectation of the system by providing him quick and accurate responses making the system more efficient and personalized to use

- *Improved Decision-Making:*

  The user he can make better decision making by providing AI tools work on giving insights about the datasets and these insights will allow the user to make better informed decision based on them such as the user will have benefit from the predictive analysis models because it gives him insights about the trend, outcome helping in achieving a better decision making.

- *Accessibility:*

  Intelligent systems can operate 24/7 there is not specific time even its features are available all this time chatbot virtual assistance.

2. Impact on Organizations

- *Customer Engagement:*

  AI powered chatbot and virtual assistants will enhance customer service and experience as we talked above about it also having his personalized experience and 34/7 customer support this will increase his loyalty to the organization system and enhance his overall engagement with the medical organization system.

- *Ethical and Regulatory Challenges:*

  Organization will ensure that there is transparency in AI decision-making processes the user will be able to know and understand the decision that are made by the organization how is it made and how it might affect them this will build trust and accountability with the system. Moreover, the company will strictly

let everyone follow laws and ethical guidelines when using AI systems to protect people's personal information and ensure fair treatment for every user this will increase the trust.

Impact of Specific Tools and Approaches

Microsoft Azure ML

Impact:

- *On Users : Azure ML might limit its accessibility for some users due to its dependency on the internet and subscription but its provides the user a user-friendly platform that can use easily without any guidance.*
- *On Organizations: Azure ML service its features are very complex to use efficiently it might require extra training for the staff but its tools are very powerful to use especially for data scientists and organization helping them to deploy the AI system more effectively .*

3. Scikit-Learn

Impact:

- *On Users: is a very user-friendly tools and very easy to use ideal for beginners but its might be restrict to use in some advanced applications such as deep learning tasks and in handling large datasets*
- *On Organizations: Scikit-Learn is a cost-effective tool for implementing machine learning models, but its limitations may require organizations to use additional tools for more complex tasks.*

4. Machine Learning (ML)

Impact:

- *On Users: ML models can provide accurate predictions and insights on the datasets, but users may* struggle to interpret complex models such as large dimensional models that need higher implementation model such as deep learning.
- On Organizations: it helps the organization to make better decision-making processes based on statistics and visualization of the data, but it might need for computational resources and expertise can increase costs so this makes it cost effective.

5. Reinforcement Learning

Impact:

- On Users: Reinforcement learning can provide adaptive and intelligent solutions that are perfect for to fulfill the user needs, but sometimes its too complex can make it hard for non-technical expertise to use it.
- On Organizations: since Reinforcement learning can offer optimizing for complex systems this will need a high computational cost and training requirements for the staff to be able to work on complex system using reinforcement learning which can be a barrier for some organizations not everyone can afford it.

Conclusion

The deployment of AI and Intelligent Systems offers transformative potential for both users and organizations and each tool, technique, type of deployment have different impact on both user and organization because each have different strengthen and limitation to know what to use this will depend on users need and organizations goals lets talk about the users benefit from what we mentioned. Users can benefit from enhanced experiences, improved decision-making, and greater accessibility, while organizations can achieve operational efficiency, innovation, and competitive advantage. However, these benefits each one comes with different challenges as we mentioned such as, including privacy concerns, ethical considerations, and the need for robust regulatory compliance. Therefore, it's very important to know what tools, techniques, approaches, and model we should select and implement depending on the application needs and goals and when we know these the organizations can maximize the positive impact of AI while mitigating potential risks.

5.Explore the technical options to enhance the performance of your model

To enhance the performance of our model, we should can consider the following technical options and apply them:

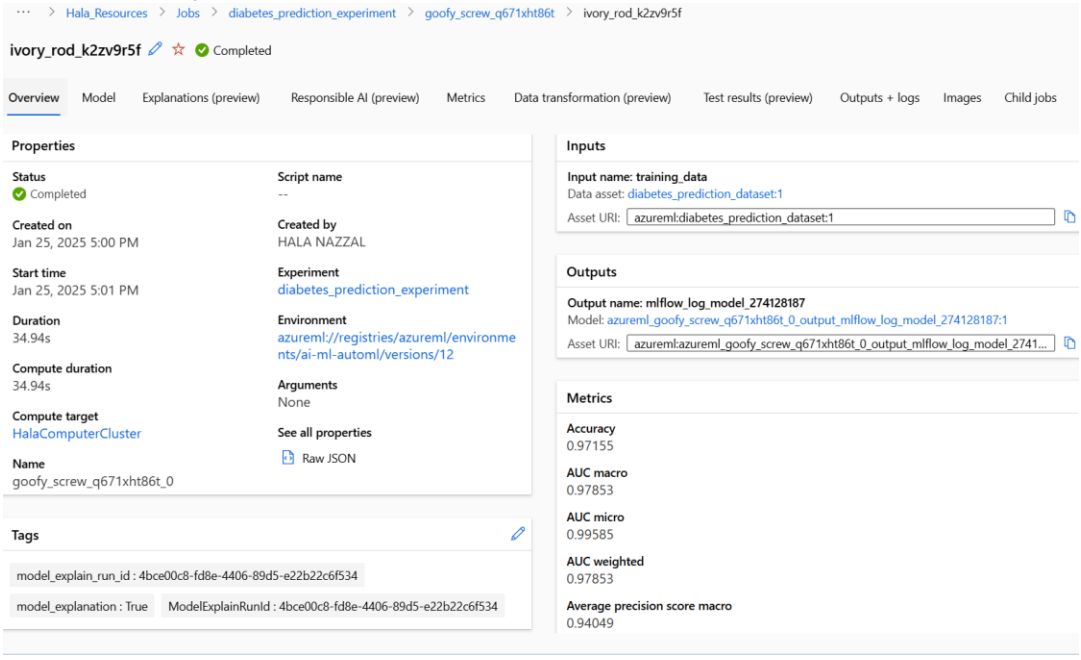
1. **Feature Engineering**:
    - **Feature Selection**: using features that will help us to having more accurate predictions that is the most relevant to the target feature that we want to predict we can know which feature is relevant enough by using correlation analysis with each feature and the target feature this is important because having un relevant feature will affect our prediction result therefor this will affect the model overall performance
    - **Feature Transformation**: normalizing the features that contains numerical value and encode the categorical is very important to make them on a similar because it'll improve model performance this can be done by label encoder , and standard scaler

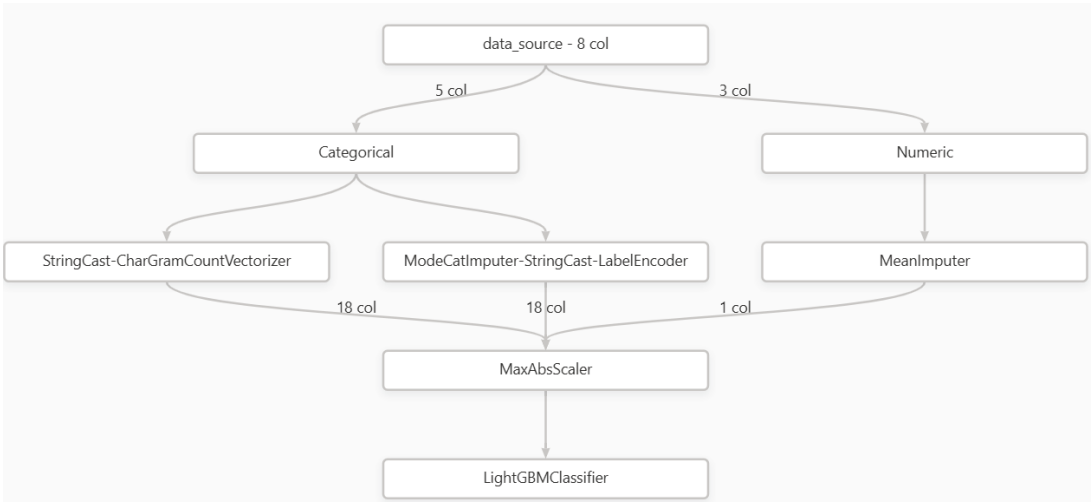
2. **Data quality:**
    - The base of the accuracy it starts with what data we have even if we used good features normalized them it doesn't mean that we have a good model so in order to have a high model accuracy we need to collect high data quality that will be

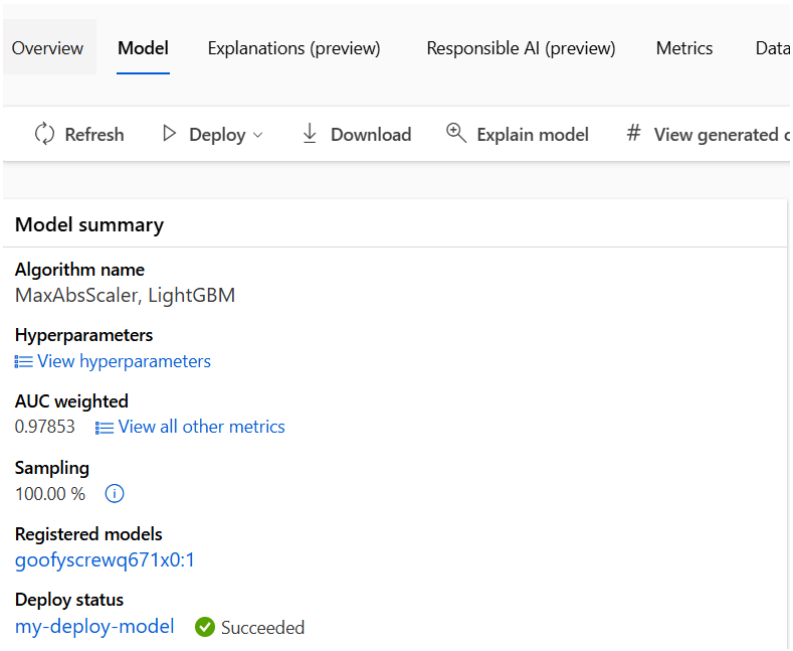
used for training it should be cleaned, well labeled, free from biases, etc. Therefore, maintaining all data conditions will result in High model quality

3. **Model Complexity:**
    - Having a complex model will cause overfitting that means when the model gets trained it will start learning from the noisy data instead of the underlying patterns. However, having a very simple model will cause underfitting which is also bad this means that there is not enough data for the model to get trained on failing to capture important nuances so, we should have balanced data that's match our task ensuring it's not too complex neither too simple.

6.Modify the implementation of the problem using Microsoft Azure and deploy it.







Here we can see how Azur ml service solved our problem first split into categorical and numerical

Used CharGramCountVectorizer and label encoder while for numerical it used meanImputer

And all of them where putted in MaxAbsScaler and inputted in LightGBMClassifier model

These were done automatically by the service itself we didn't need to code anything

7.Demonstrate your modification and show enhancement using benchmarking metrics.

The Azure ml can display different metric charts as we can see here, he displayed a lot of metrics

Such as confusion matrix, accuracy table precision-recall ,roc curve, Calibration curve, lift curve

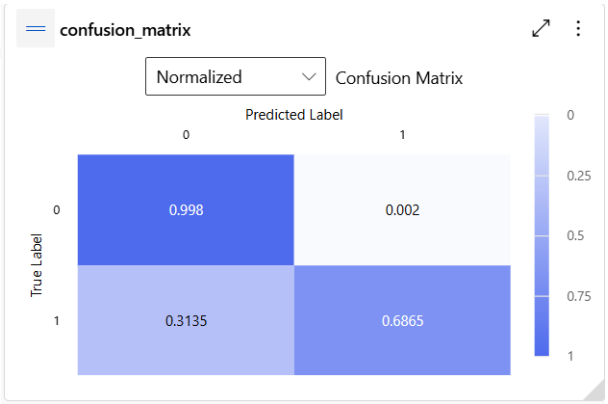
With information on each chart this is done automatically without coding this will enable the staff see it quicker without the need for a programmer to code it.

Confusion matrix Help the user to visualize true positives, true negatives, false positives, and false negatives.

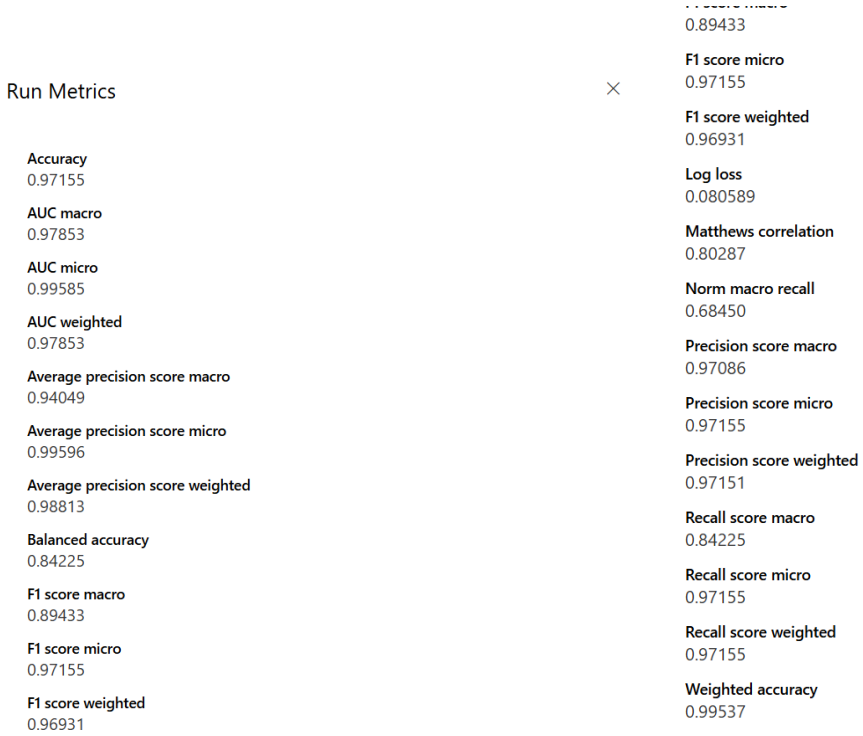
```
Accuracy: 0.96
Confusion Matrix:
[[18064   158]
 [  653  1122]]
Classification Report:
              precision    recall  f1-score   support

           0       0.97      0.99      0.98     18222
           1       0.88      0.63      0.73      1775

    accuracy                           0.96     19997
   macro avg       0.92      0.81      0.86     19997
weighted avg       0.96      0.96      0.96     19997
```



1. Confusion matrix in azure
    - Accuracy: the overall accuracy of the model is 0.97155
    - **Confusion Matrix:** Provides raw counts of true and predicted labels.
        - **True Label 0:** 91,320 (predicted as 0) and 2,665 (predicted as 1).
        - **True Label 1:** 5,835 (predicted as 0) and 180 (predicted as 1).
2. Confusion matrix in colab:
    - **Accuracy**: The overall accuracy of the model is 0.96.
    - **Confusion Matrix**: This is a raw confusion matrix.
        - True Label 0: 18,064 (predicted as 0) and 158 (predicted as 1).
        - True Label 1: 653 (predicted as 0) and 1,122 (predicted as 1).
    - **Classification Report**:
        - **Precision, Recall, F1-Score**:
            - Class 0: Precision = 0.97, Recall = 0.99, F1-Score = 0.98.
            - Class 1: Precision = 0.88, Recall = 0.63, F1-Score = 0.73.

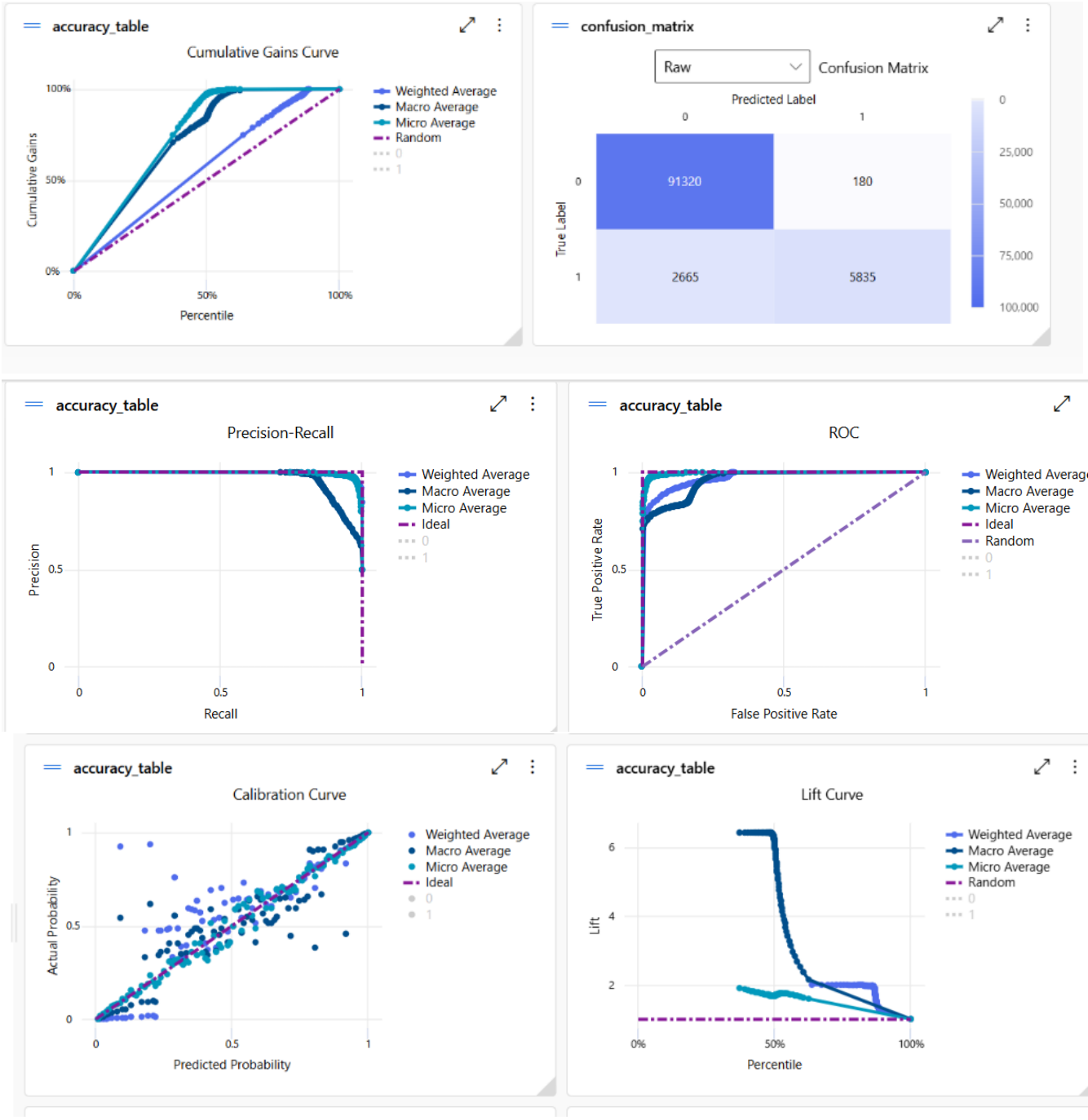


F1 score for macro is also higher by 0.03 here its 0.89433

As we can see here the weighted accuracy here is higher than Colab here its 0.99537

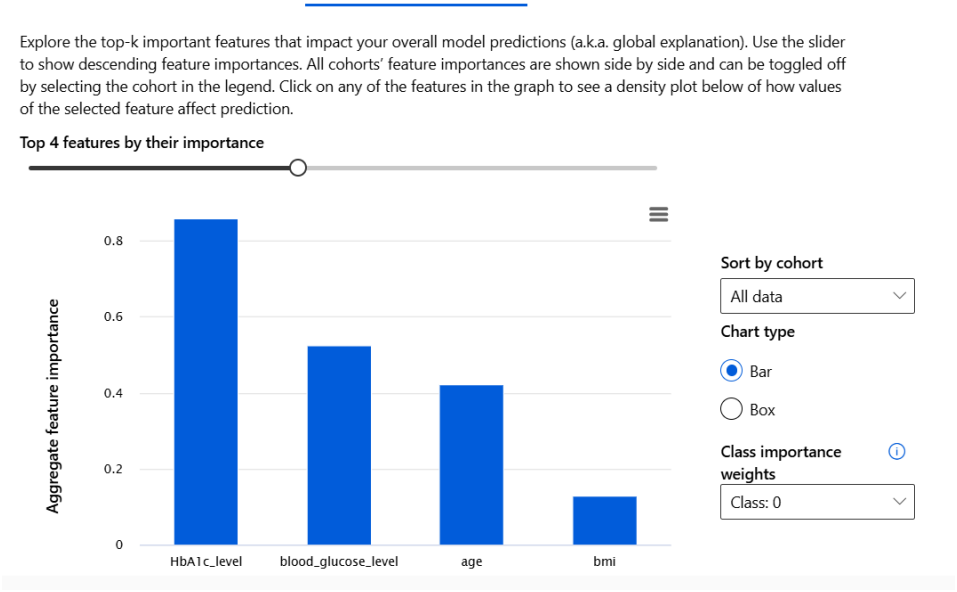
Summary:

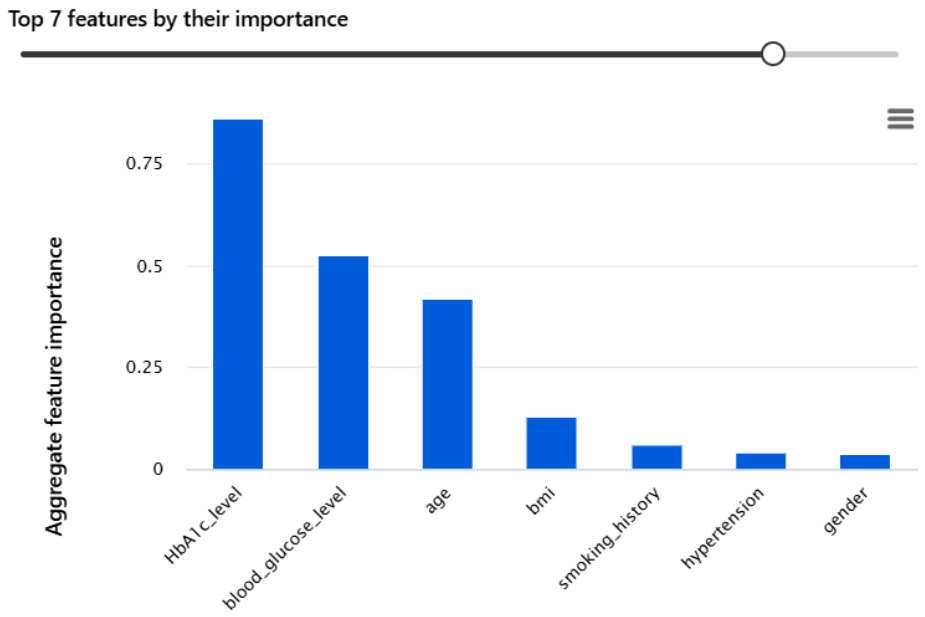
- Azure ML provides various metrics and table these will help us the hospital manager to visualize and understand the model's performance with analyzing what are the true positives, true negatives, false positives, and false negatives and these will enable them to see the difference between Azure ML and Colab, showing that by having all of these results that using Azure ML will achieve higher accuracy and F1 scores.



Azure can help us to see what are the features that most relevant features to the model performance as we can see these are first 4 features that are most relevant to the model without the need for coding

Make it quicker





Here in this graph, we can see the most to the least relevant feature as we can see here the gender was the least important feature, we can also see what's the value of importance.

8.Evaluate your own role to improve the performance of your model. (Ranta, 2024) (ovic, n.d.)

My role I is to improve the performance of the diabetes prediction model and the first step to do that is to involve a combination of technical expertise and soft skills in the system to make sure that the system is accurate, efficient, and aligned with organization goals also in user needs.

We have many stages to improve the performance model level lets start with the first stage which is data collection and cleaning phase, in this stage I gathered comprehensive patient data that are needed in the prediction such as BMI, HbA1c, age, and blood glucose levels, then we ensured the our data that we gathered clean and what do we mean by cleaned data it means that its doesn't contains null values, well-labeled, free from nosy data. Then I handled missing values if there is any using techniques imputations (filling with mean , mode, median).Additionally, I checked if the data that we collected is it free from biases that could skew our model's prediction and we made sure that our data is representative. The second stage, I went to feature engineering stage I included normalizing and label encoding, normalizing was used for the numerical features by standard scaler to ensure that they are on a similar scale then we label encoded using label encoder to make them suitable for the machine learning algorithms this step is crucial to make since it helps the model's ability to identify a meaningful relationship with the diabetes risk column since it's the target feature. In the third stage training and optimization for the model happens, I used Azure ML and used logistic regression model to train diabetes dataset we optimized model performance by using se accuracy, confusion matrix, and classification report to see how well your model is performing and where it might be making mistakes. In the fourth stage I focused on the Interpretability so, I can understand how my model is working what are the important features that are might affect the overall performance of our model what are the relevant features that we have I used technique called coefficient analysis helped me to see the relations. Moreover, to monitoring of the model's performance across various iterations I decided to use ML flow and to improve the model accuracy I decided to combine fuzzy logic mathematical approach and logistic regression because fuzzy logic help us to fix any uncertainties and imprecision in the data enhancing so, this hybrid approach will enhance our model's ability to predict diabetes risk by expert knowledge through fuzzy rules with strong statistical implementation by logistic regression. Therefore, by adding MLflow for experiment tracking and integrating fuzzy logic with logistic regression, the model's performance, interpretability, and management will be significantly improved, aligning with organizational goals and user needs. In The last stage, I used Azure ML service when it comes to deployment and to compare the deployment that we had in colab to see which one is better for our system, the deployment will be able to provide insights, visualization, statistical analyses, streamlines the evaluation and monitoring of model performance and more these will be shown automatically without the need to type any code also it provides an virtual unlimited storage and strong computational resource to use ensures that the deployment can handle large datasets and high-demand scenarios efficiently.

Finally, I prioritize ethical considerations throughout the entire process. I ensure the model is doesn't contain any from biases I also focused on transparency and explainability made sure that everything is clear how was the decision making is made and where the data used of the patient also for what reason is it used for. Additionally, I strictly agree to healthcare regulations and ethical guidelines, such as HIPAA, that help in protecting the patient data ensure compliance.

To conclude, I combined technical skills in data preprocessing, feature engineering, model optimization, and deployment with soft skills like ethical awareness this will ensure that my diabetes prediction model isn't only accurate but will also align with the organizations goals and user needs because its trustworthy. This holistic approach ultimately delivers value for both users and the organization, improving healthcare outcomes and enhancing decision-making processes.

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